INTERNATIONAL MONETARY FUND AND
INTERNATIONAL DEVELOPMENT ASSOCIATION

BENIN

Joint IMF/World Bank Debt Sustainability Analysis 2007[1]

Prepared by the staffs of the International Monetary Fund and
the International Development Association

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Benin's risk of external debt distress is moderate*. All external debt indicators in the baseline scenario are far below the relevant policy-dependent indicative thresholds. However, the NPV of debt-to-export threshold is breached under scenarios incorporating lower growth, more volatile cotton production and less generous financing terms. The fiscal sustainability analysis confirms these results and highlights concerns about the fiscal cost of large domestic borrowing with high interest rates and shorter maturities. The authorities could reduce the risks associated with this moderate rating and safeguard debt sustainability through sustained efforts to continue structural reforms (including measures to spur private sector participation in development), address absorption constraints, and pursue prudent debt management policies*.

I. INTRODUCTION

1. This note, which updates the November 2006 DSA (Country Report 07/6), projects a lower debt-to-GDP ratio at the end of 2007 than in the 2006 exercise, reflecting three main

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[1] Prepared jointly by the IMF and World Bank staffs using the Debt Sustainability Framework for low-income countries. The analysis is based on Beninese authorities' public debt management agency (*Caisse Autonôme d'Amortissement)* debt database and on IMF and World Bank staff estimates.

factors: (i) the full impact of the Multilateral Debt Relief Initiative (MDRI) from the World Bank (US$664 million, against an earlier estimate of US$632 million) and the African Development Bank (US$364 million against US$296 million); (ii) the 2007 cancellation of Benin's bilateral debt both to China (US$23.2 million) and Russia (US$11.6 million); and (iii) the addition of two publicly guaranteed commercial loans totaling US$56.6 million (Text table 1).

Text table 1. Benin: Differences Underpining the DSA 2006 and 2007
(Percent, as of end-2006)

	2007 DSA
Debt-to-GDP ratio	12.0
2006 DSA	13.0
Update of MDRI relief 1/	-1.6
Other bilateral cancellations 2/	-0.6
Debt guarantees 3/	1.2

Source: Beninese authorities and staff estimates.

1/ Effective contribution of IDA and ADF.
2/ Additional cancellations from China and Russia.
3/ The 2007 DSA integrates commercial debt that is publicly guaranteed.

2. Recent debt relief—under the Heavily Indebted Poor Countries Initiative in 2003 and MDRI in 2006—have greatly reduced Benin's debt stock and notably changed its structure. The external public debt-to-GDP ratio fell to 12 percent in 2006, from about 59 percent in 2001 (before HIPC relief). Debt to Paris Club creditors (including Russia) has been cancelled. While the combined share of commitments to the three major multilateral creditors (the IDA, AfDB, and the IMF) has declined from about 71 percent to 36 percent, the portion of debt to other multilateral creditors has risen to 44 percent, from 13 percent before HIPC/MDRI.[2]

[2] Other multilateral institutions include the West African Development Bank (14.8 percent of total debt), the Islamic Development Bank (8.8 percent of total debt), the European Investment Bank (5.1 percent of total), and the Arab Bank for Economic Development in Africa (0.6 percent of total).

Text table 2. Benin: External Public and Publicly Guaranteed Debt Structure (As of end-2006)

	Before HIPC relief		After HIPC relief 1/		Post MDRI		
	Millions of US$	Share in percent	Millions of US$	Share in percent	Millions of US$	Share in percent	Percent of GDP
Multilateral 3/	1,731.5	84.5	1,539.8	93.2	454.8	80.1	9.6
World Bank	863.3	42.1	779.7	47.2	115.6	20.4	2.4
African Fund for Development	520.3	25.4	451.4	27.3	87.3	15.4	1.8
IMF	65.2	3.2	59.4	3.6	2.7	0.5	0.1
Others	282.8	13.8	249.3	15.1	249.3	43.9	5.2
Bilateral	260.7	12.7	56.1	3.4	56.1	9.9	1.2
Paris Club	189.0	9.2	0.0	0.0	0.0	0.0	0.0
Non-Paris Club 2/	71.8	3.5	56.1	3.4	56.1	9.9	1.2
Commercial	56.6	2.8	56.6	3.4	56.6	10.0	1.2
Total	2,048.9	100.0	1,652.5	100.0	567.5	100.0	12.0

Source: Beninese authorities.
1/ At end-2006 with HIPC relief obtained since July 2000.
2/ All non-Paris Club creditors have provided comparable treatments except Libya, Nigeria and the DRC.

II. External Debt Sustainability Analysis

A. Baseline Scenario: 2007–27[3]

3. The baseline macroeconomic scenario projects a gradual strengthening of real GDP growth, a steady increase in private investment, and a stable inflation outlook. Growth is projected to pick up from 4.2 percent in 2007, to 5.7 percent by 2010 (in CFA francs, see Text table 3), and average 6 percent annually in 2011–27. The projection assumes that key structural reforms and infrastructure improvements are gradually carried out (Country Report 07/213). The cotton industry is expected to reach its full ginning capacity of 600,000 tons by 2027, and long-run growth, underpinned by private-sector-led economic diversification, should be sustained. Private investment is projected to grow a steady 0.1–0.2 percentage point of GDP a year, reaching 17.3 percent of GDP by 2027, up from 13.6 percent in 2006. Inflation should stay below the 3 percent ceiling set by the West African Economic and Monetary Union's (WAEMU).

[3] See Box 1 for more details on the baseline assumptions.

Text table 3. Benin: 2006 and 2007 DSA Comparative Assumptions
(Percent, unless otherwise indicated)

	DSA 2007					DSA 2006				
	2006	2007	2008	2009	2010-27	2006	2007	2008	2009	2010-26
	Act.	Projection				Projection				
Real GDP growth (CFA)	3.8	4.2	5.4	5.7	6.0	4.5	5.1	5.5	5.5	5.5
PRSP baseline real GDP growth	3.8	5.3	7.0	7.5	10.1	4.5	6.0	6.8	7.5	10.2
GDP deflator (CFA)	3.1	3.0	2.8	2.8	2.9	2.6	2.4	2.4	2.4	2.4
Real export growth	-13.8	18.1	13.3	5.4	9.6	0.7	21.3	10.2	15.2	11.9
Fiscal revenue (percent of GDP)	16.8	19.0	18.5	18.8	19.1	16.5	17.2	18.0	18.1	18.9

Sources: Beninese authorities; and IMF staff estimates and projections.

4. The 2007 DSA baseline is built on an updated macroeconomic scenario. Real GDP growth, projected to be 1.4 percentage point above Benin's historical average, assumes that key structural reforms and infrastructure improvements are gradually carried out (Country Report 07/213). Also, with more ambitious capacity increases at the Port of Cotonou, and more sustained public works activities than earlier anticipated, baseline long-term growth is 0.5 percentage-point higher than assumed in the 2006 DSA.

Box 1. Benin: Baseline Scenario—Macroeconomic Assumptions for 2006–27
(Estimates and projections are in CFA franc terms, unless otherwise indicated)

Annual real GDP is projected to rise by 5.4 percent in 2007–12 and by 6 percent in 2013–27 (see paragraph 3 and 4 above, for more details).

Inflation, as measured by the consumer price index, is projected to fall gradually from a historical average of 3.2 percent in 2000–06, to below 3 percent thereafter, in line with the relevant WAEMU convergence criterion.

Fiscal performance is projected to remain strong, reinforced by debt relief savings because of strong revenue collection and prudent expenditure management. The overall fiscal deficit, excluding grants, is projected to remain below 5 percent of GDP.

Against a background of improving terms of trade and strengthened domestic policy and reform implementation, **the trade deficit** would narrow marginally, from 10 percent in 2006 to 9 percent by 2027.

The **financing gap** is expected to decline gradually from 1.5 percent of GDP in 2008 to 1.2 percent in 2010 and onward, 80 percent of which is assumed to be covered by **grants**. Furthermore, **debt-creating flows** that include both identified projects and program loans are projected to average 3 percent of GDP.

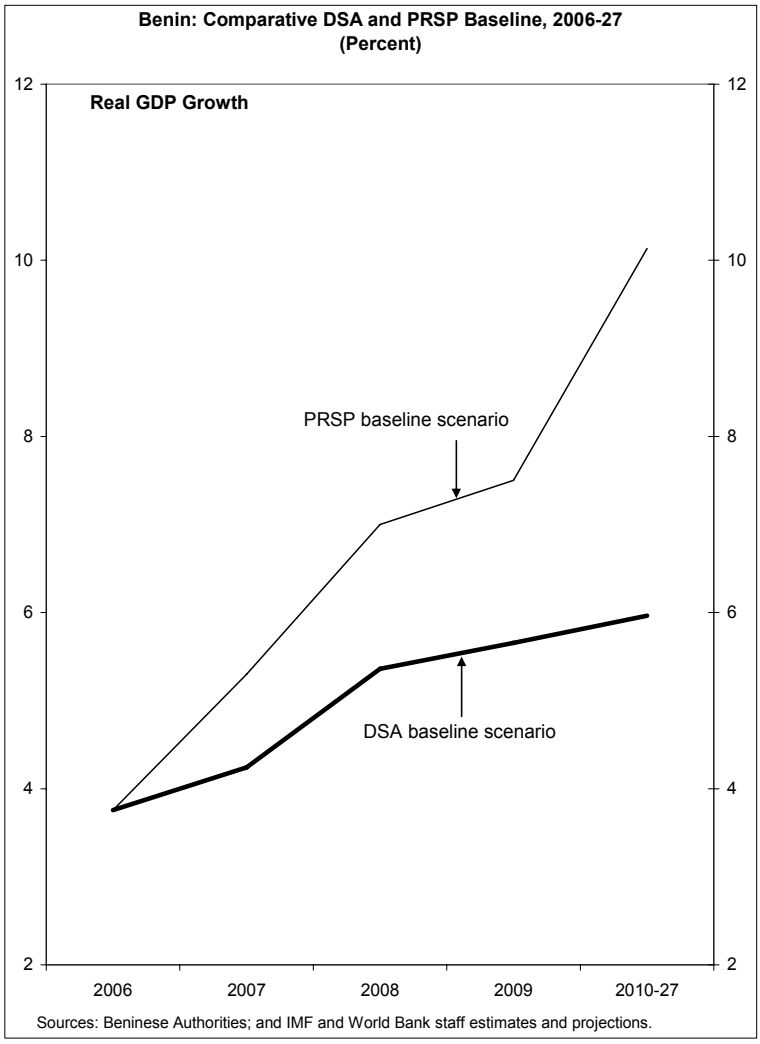
As a result, new public borrowing (residual financing gap plus debt-creating flows) is projected to decline from 2.9 percent of GDP in 2007 to 2.4 percent thereafter. Post-MDRI distribution of new loans among creditors would be as indicated in the box table below. With a projected average maturity of 31 years, a grace period of 6 years, and an interest rate of 2.4 percent; and assuming an average discount rate of 5 percent, the new loans are estimated to have an average grant element of about 30.7 percent.

Box table. Benin: Assumed Shares of New Loans by Creditor, 2006–27 [1/]
(Percent)

	2006	2007	2008	2009	2010	2011-27
Multilaterals	80.1	78.3	76.5	74.7	72.9	73.9
IMF	0.5	0.5	0.5	0.5	0.5	0.5
IDA	20.4	20.6	20.8	21.0	21.2	22.2
Others	59.3	57.3	55.3	53.3	51.3	51.3
Official Bilaterals	9.9	12.2	14.5	16.8	19.1	22.6
Paris Club	0.0	2.3	4.6	6.9	9.2	12.7
Non-Paris Club	9.9	9.9	9.9	9.9	9.9	9.9
Commercial	10.0	9.5	9.0	8.5	8.0	3.5

Source: Staff estimates and projections as of 2007.
1/ Projection as of 2007.

5. The baseline (average) growth scenario is realistic given the country's structural constraints. It assumes full implementation of medium-term structural reforms with moderate gains in long-run growth (2011–27). Growth would not deviate considerably from recent trends owing to residual institutional and structural weaknesses (e.g., in the cotton sector, the judiciary, and the business environment). Scarcity of land and modest growth of private investment are also likely to limit long-run growth.



6. The scenario assumes continued strong revenue collection and further budget consolidation. Public expenditure is projected to increase steadily by 0.5 percentage point a year, to about 23 percent of GDP in 2010, from 19 percent in 2007. Higher revenue would partially help meet the projected scaling-up of public expenditure, and the average overall fiscal deficit, excluding grants, would stay below 5 percent of GDP.

B. External Debt Sustainability[4]

7. The DSA results point to a moderate risk of debt distress for Benin. As shown in Appendix Figure 1, all debt indicators remain well below their respective policy-dependent

[4] The latest World Bank Country Policy and Institutional Assessment classifies Benin as a "medium performer," which puts the debt burden thresholds for the country are as follows: (i) debt to export of 150 percent (NPV); (ii) debt to GDP of 40 percent (NPV); (iii) debt to fiscal revenue of 250 percent (NPV); (iv) debt service to export of 20 percent; and (v) debt service to fiscal revenue of 30 percent.

thresholds under the baseline scenario. However, in bound tests, the threshold for the NPV of the external debt-to-exports ratio is breached when (i) real GDP grows at its historically low average rate (which could happen if reforms either lag or cease); (ii) cotton production and exports are lower than in the baseline; and (iii) nonconcessional borrowing is undertaken. Furthermore, relevant stress tests indicate that nonconcessional borrowing or exchange rate shocks could cause debt ratios to approach, but not exceed, some of the indicative debt thresholds. These findings are consistent with the staffs' 2006 DSA.

Text table 4. Benin: External Debt Indicators after MDRI, 2006–27
(Percent, unless otherwise indicated)

	Indicative thresholds	Baseline Scenario					
		2006	2007	2008	2009	2010	2011–27 1/
		Act.	Projection				
NPV of debt to GDP	40.0	8.8	9.2	9.7	10.2	10.6	13.2
Debt service to revenue	30.0	3.8	3.6	3.7	3.4	3.5	4.4
NPV of debt to revenue	250.0	47.9	48.6	52.5	54.3	55.3	68.7
Debt service to exports	20.0	6.3	6.0	5.5	5.3	5.5	6.5
NPV of debt to exports	150.0	75.2	80.8	79.6	84.3	88.8	101.6
Memorandum items:							
Debt to GDP (post-HIPC)		36.0	34.1	34.5	35.4	35.9	34.3
Debt to GDP (post-MDRI)		12.0	13.6	14.2	14.9	15.4	18.4

Sources: Beninese authorities; and IMF staff estimates and projections.
1/ Simple average.

III. Fiscal Sustainability Analysis[5]

8. Domestic borrowing—notwithstanding the potential space created by debt relief and the easing of WAEMU restrictions on borrowing on the regional financial market—is expected to increase at a modest pace. This reflects the country's broadly favorable medium-term fiscal prospects as well as delays in addressing absorption constraints. Domestic debt rose markedly in 2006–07, from about 0.6 percent to 4.0 percent of GDP, largely because of the central government's issue of treasury bonds on the regional financial market, which amounted to CFAF 96.1 billion (3.7 percent of GDP).[6] By 2027, domestic debt is projected to increase to 10 percent of GDP (see Appendix Figure 2).

[5] Domestic debt (including borrowing from the WAEMU regional market) is likely to become a policy issue requiring close monitoring. Staff projects such debt to rise to 9–10 percent of GDP by 2027, from about 3.5 percent of GDP in 2007.

[6] This concerns all WAEMU residents who hold Beninese central government Treasury bonds or bills.

9. The fiscal cost associated with accelerated domestic borrowing could eventually test budget sustainability, as is the case under a scenario where reforms stagnate, domestic borrowing at high interest rate and shorter maturities surges, or institutional weaknesses in the cotton sector continue.

10. Continued fiscal discipline would help Benin weather economic shocks and keep its public debt sustainable (see Box 2). Even with good fiscal performance, excessive nonconcessional borrowing and other persistent shocks could require Benin to make fiscal adjustments in the future to keep debt dynamics from deteriorating severely. The average fiscal saving required to keep debt sustainable in alternative DSA scenarios are as follows:

- In the *increasing domestic(including regional) borrowing* scenario, the average fiscal saving required to keep the debt ratio from rising (relative to the stabilizing primary balance scenario) is about 3.3 percent of GDP a year.

- In the *no reform* (status quo) scenario, Benin would need fiscal saving of 3.1 percent of GDP on average a year.

- In the "*instability in the cotton sector*" shock, the required savings is estimated at 3.5 percent of GDP per year.

Box 2. Debt Dynamics and Fiscal Stance

We use standard debt dynamics equations that link the fiscal stance to the debt ratio to answer the following two questions (see Appendix C for more detail):

(i) What ratio of primary fiscal surplus is required to stabilize the public debt (ratio to GDP) at its base year level if policy remains unchanged?

(ii) What primary surplus is associated with (a) a no-reform scenario? (b) a sharp increase in a government T-bill-financed fiscal deficit issued in the regional market? or (c) a non-broad-based growth scenario that assumes continued volatility in both cotton production and international cotton prices.

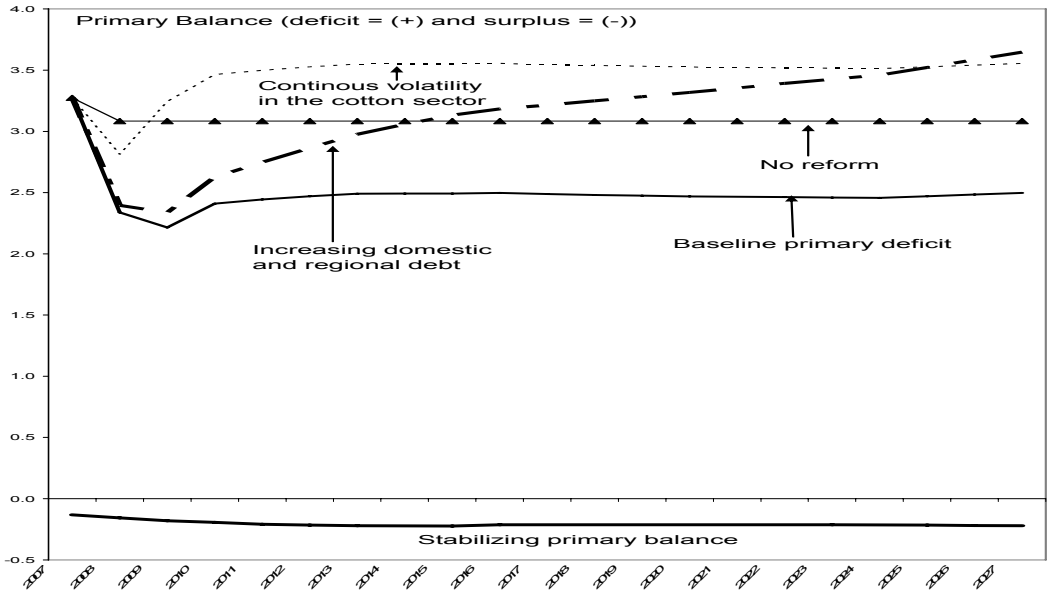
The figure below suggests that:

- An average fiscal surplus of 0.2 percent of GDP (per year in 2007-27) is required to bring the ratio of public debt–to-GDP to its 2006 level of 14.7 percent.

- The primary deficit is higher in the no-reform scenario than in the baseline; the weak growth and cotton sector performance, and increasing domestic debt scenarios would result in a much bigger fiscal deficit.

- The fiscal saving required to keep the debt ratio from rising relative to the stabilizing primary balance scenario ranges from 3.1 to 3.5 percent of GDP a year.

Box Figure. Benin:Primary Balance Under Three Shock Scenarios and Required Primary Balance to Stabilize The Debt Ratio, 2007-27 (in percent of GDP)



Source: IMF staff projections and simulations.

IV. CONCLUSION

11. IMF and World Bank staffs conclude that Benin faces a moderate risk of debt distress. All external debt indicators in the baseline scenario are far below the relevant policy-dependent indicative thresholds. However, the NPV of debt-to-export (indicative) threshold is breached under some alternative scenario and bound tests.

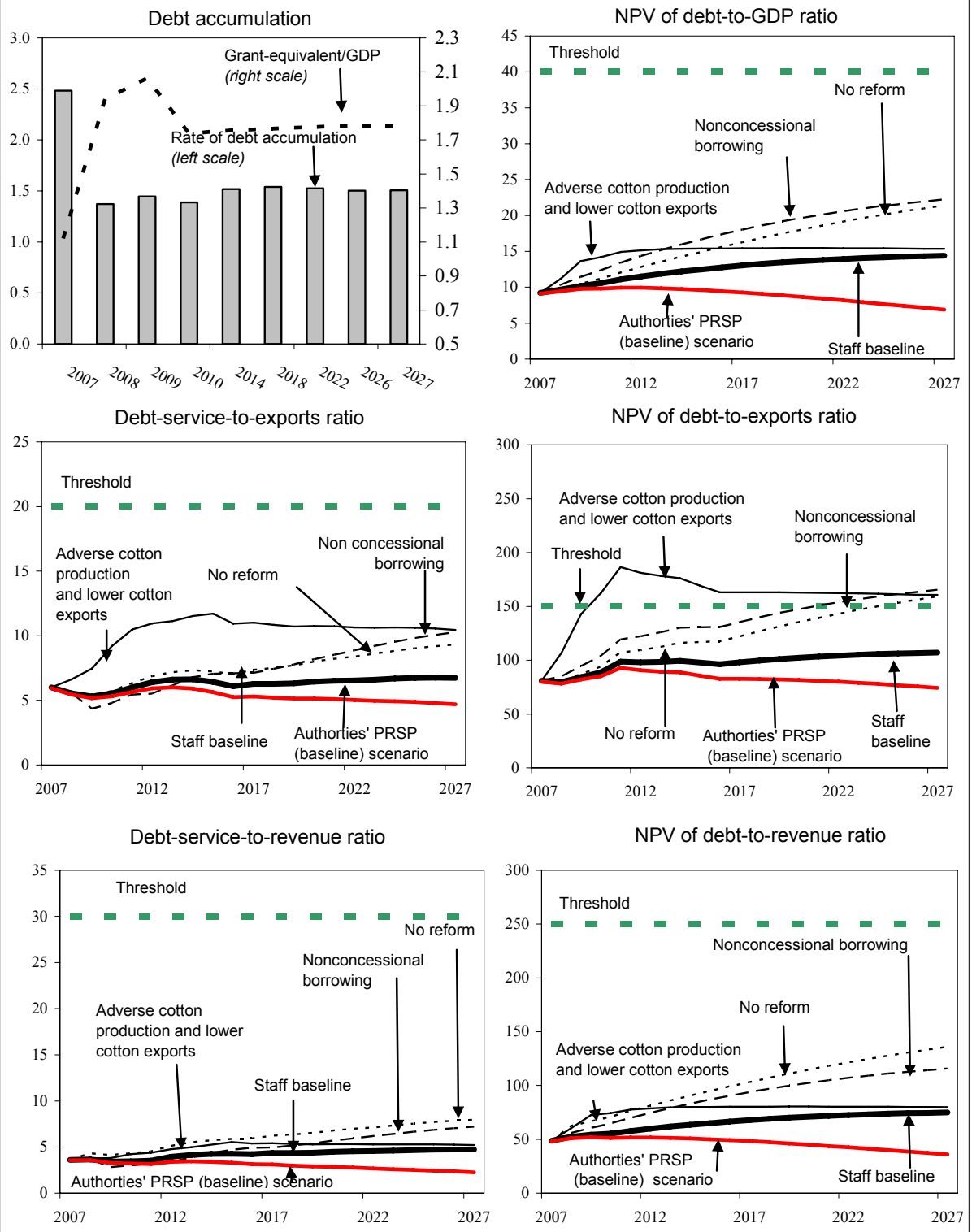
12. The key risks driving the moderate rating are as follows: (i) weak growth; (ii) cotton production and exports being lower than in the baseline; and (ii) nonconcessional borrowing. The authorities could reduce these risks and safeguard debt sustainability through sustained efforts to continue structural reforms (including measures to spur private sector participation in development), address absorption constraints, and pursue prudent debt management policies.

13. Excessive domestic borrowing at high interest rates and shorter maturities would require Benin to make substantial fiscal adjustment down the road to keep debt dynamics from deteriorating. Continued fiscal consolidation and fiscal discipline would, therefore, reduce the potential for fiscal imbalances that put macroeconomic stability at risk.

14. The Beninese authorities concurred broadly with the DSA findings. While recognizing the need to refrain from nonconcessional borrowing, the authorities pointed to difficulties in mobilizing grant and concessional resources, which in their view could slow Benin's progress toward the Millennium Development Goals.

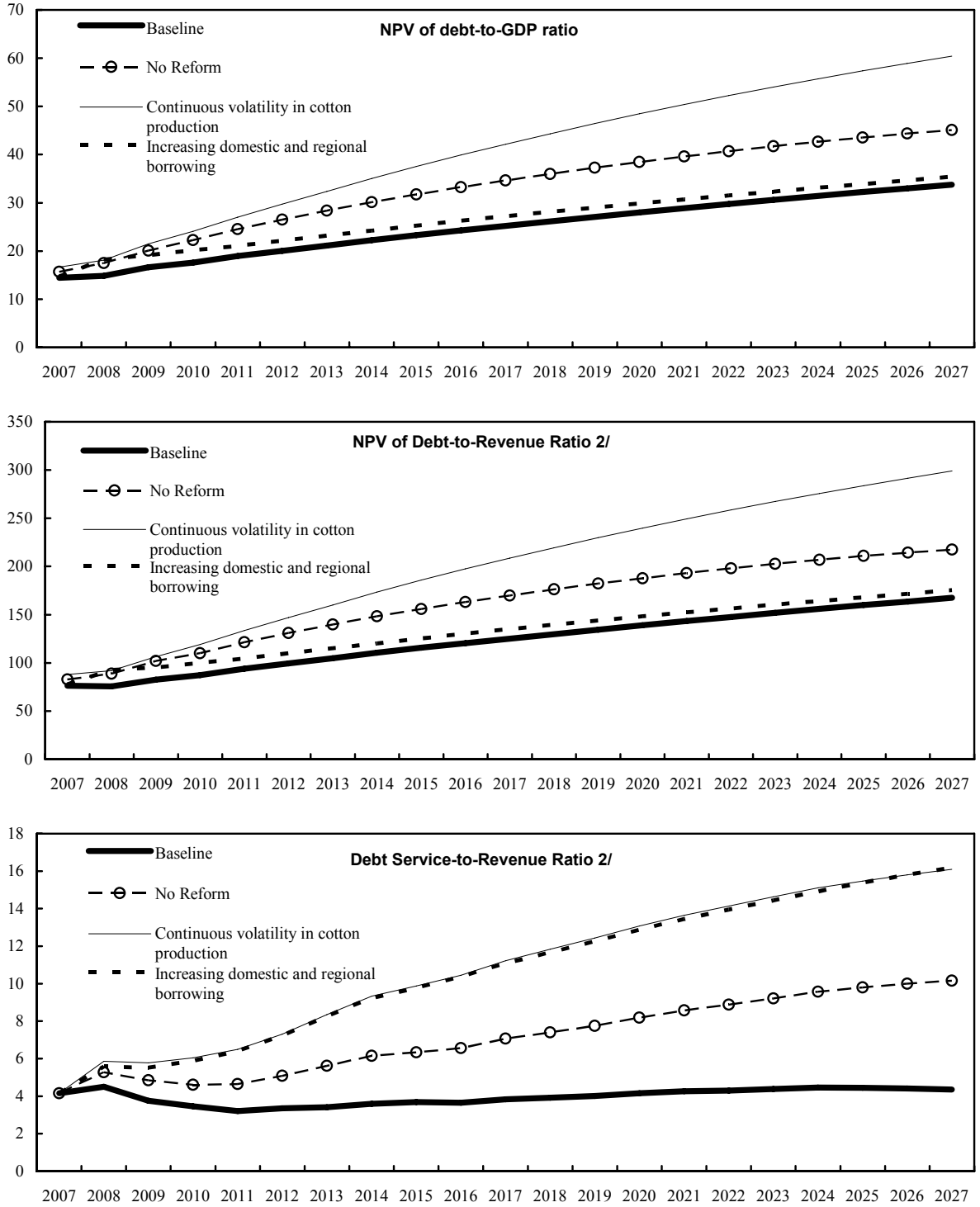
Appendix A: Figures

Appendix Figure 1. Benin: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2007–27



Sources: IMF and World Bank staff projections and simulations.

Appendix Figure 2. Benin: Indicators of Public Debt Under Alternative Scenarios, 2007–27 1/



Sources: IMF and World Bank staff projections and simulations.
1/ Most extreme stress test is that which yields highest ratio in 2017.
2/ Revenue including grants.

Appendix Table 1. Benin: External Debt Sustainability Framework, Baseline Scenario, 2004–27 1/
(Percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Projections									
	2004	2005	2006	Average 6/	Deviation 6/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
External debt (nominal) 1/	33.8	36.9	12.7			14.2	14.7	15.3	15.7	16.2	16.6		18.3	19.8	
Of which: Public and publicly guaranteed (PPG)	33.8	36.9	12.0			13.6	14.2	14.9	15.4	16.1	16.5		18.3	19.8	
Change in external debt	-2.7	3.1	-24.2			1.5	0.5	0.6	0.4	0.5	0.4		0.3	0.1	
Identified net debt-creating flows	1.6	1.0	1.3			3.4	4.3	4.2	4.0	3.9	3.8		3.5	2.8	
Non-interest current account deficit	6.8	5.6	5.9	6.4	1.1	5.4	5.8	5.7	5.7	5.6	5.6		5.4	5.2	5.4
Deficit in balance of goods and services	12.2	9.8	11.5			11.5	11.0	10.7	10.6	10.6	10.6		10.4	10.2	
Exports	14.3	13.0	10.7			11.4	12.2	12.1	11.9	11.3	11.8		13.3	13.5	
Imports	26.6	22.8	22.2			22.9	23.1	22.8	22.5	21.9	22.3		23.8	23.7	
Net current transfers (negative = inflow)	-6.0	-4.8	-5.7	-5.7	0.6	-6.3	-5.4	-5.3	-5.2	-5.2	-5.2		-5.2	-5.2	-5.2
Of which: Official	-3.2	-2.0	-2.5			-3.1	-2.4	-2.3	-2.3	-2.3	-2.4		-2.6	-3.2	
Other current account flows (negative = net inflow)	0.5	0.5	0.1			0.2	0.2	0.2	0.3	0.2	0.2		0.2	0.2	
Net FDI (negative = inflow)	-1.2	-2.2	-2.3	-1.5	0.8	-1.1	-0.9	-0.9	-1.0	-1.1	-1.1		-1.2	-1.5	-1.3
Endogenous debt dynamics 2/	-4.0	-2.4	-2.3			-0.9	-0.5	-0.5	-0.6	-0.7	-0.7		-0.8	-0.8	
Contribution from nominal interest rate	0.4	0.3	0.3			0.4	0.3	0.3	0.3	0.3	0.3		0.4	0.4	
Contribution from real GDP growth	-4.3	-1.0	-1.6			-1.2	-0.8	-0.8	-0.9	-1.0	-1.0		-1.1	-1.2	
Contribution from price and exchange rate changes	-0.1	-1.7	-1.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	-4.3	2.2	-25.5			-1.9	-3.8	-3.7	-3.6	-3.4	-3.4	-3.3	-3.2	-2.8	-3.1
Of which: Exceptional financing	-1.7	-0.9	-2.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	...	...	8.8			9.9	10.2	10.6	10.8	11.3	11.6		13.0	14.4	
In percent of exports	...	...	82.2			86.6	83.7	87.3	90.9	100.0	98.6		97.9	107.1	
NPV of PPG external debt	...	...	8.1			9.2	9.7	10.2	10.6	11.1	11.5		13.0	14.4	
In percent of exports	...	...	75.2			80.8	79.6	84.3	88.8	98.6	98.0		97.9	107.1	
In percent of government revenues	...	...	47.9			48.6	52.5	54.3	55.3	57.8	60.0		67.8	75.0	
Debt service-to-exports ratio (percent)	10.6	8.9	6.3			6.3	6.7	6.3	6.4	6.8	7.1		6.3	6.7	
PPG debt service-to-exports ratio (percent)	10.6	8.9	5.9			6.0	5.5	5.3	5.5	6.0	6.4		6.3	6.7	
PPG debt service-to-revenue ratio (percent)	9.2	7.0	3.8			3.6	3.7	3.4	3.5	3.5	3.9		4.3	4.7	
Total gross financing need (billions of U.S. dollars)	0.3	0.2	0.2			0.3	0.3	0.4	0.4	0.4	0.5		0.7	1.6	
Non-interest current account deficit that stabilizes debt ratio	9.5	2.4	30.1			3.8	5.3	5.1	5.3	5.1	5.2		5.2	5.1	
Key macroeconomic assumptions															
Real GDP growth (percent, U.S. dollar)	13.4	3.1	4.6	4.7	9.6	11.1	6.3	6.2	6.4	6.7	6.7	7.2	6.8	6.9	6.8
GDP deflator in US dollar terms (change in percent)	0.3	5.4	3.1	2.9	1.8	3.0	2.8	2.8	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Effective interest rate (percent) 5/	1.3	1.1	1.0	1.4	0.3	3.2	2.3	2.2	2.1	2.1	2.1	2.3	2.1	2.2	2.1
Growth of exports of G&S (U.S. dollar terms, percent)	19.2	-1.6	-11.1	2.8	14.5	21.8	16.6	8.6	7.8	3.7	14.7	12.2	10.0	10.1	10.9
Growth of imports of G&S (U.S. dollar terms, percent)	13.9	-6.8	5.2	5.7	9.8	17.9	10.3	7.7	8.2	6.5	12.2	10.5	9.8	9.9	10.3
Grant element of new public sector borrowing (percent)	...	...	...	...	...	29.8	29.7	29.7	29.7	30.1	30.2	29.9	30.7	31.6	31.0
Aid flows (billions of U.S. dollars) 7/	0.0	0.0	0.0			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.4	
Of which: Grants	0.0	0.0	0.0			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.4	
Of which: Concessional loans	0.6	0.7	0.8			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (percent of GDP) 8/	...	...	...			1.1	1.9	2.1	1.7	1.8	1.8		1.8	1.8	1.8
Grant-equivalent financing (percent of external financing) 8/	...	...	...			29.8	52.8	54.2	51.0	51.3	51.4		51.7	52.3	51.9
Memorandum items:															
Nominal GDP (billions of U.S. dollars)	4.1	4.4	4.7			5.4	5.9	6.5	7.1	7.8	8.6		13.7	35.2	
(NPVt-NPVt-1)/GDPt-1 (percent)						2.5	1.4	1.4	1.4	1.6	1.5	1.6	1.5	1.5	1.5

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate, g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief), changes in gross foreign assets, and valuation adjustments. Projections also include contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Appendix Table 2. Benin: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of debt-to-GDP ratio								
Baseline	9.2	9.7	10.2	10.6	11.1	11.5	13.0	14.4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	9.2	9.8	10.5	11.2	12.1	12.8	16.3	21.4
A2. New public sector loans on less favorable terms in 2008-27 2/	9.2	10.3	11.4	12.4	13.4	14.4	18.0	22.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	9.2	10.7	12.3	14.1	16.4	18.8	31.6	35.0
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	9.2	11.1	13.6	14.2	14.9	15.1	15.4	15.4
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008-09	9.2	9.8	10.5	11.1	11.9	12.3	13.9	15.4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	9.2	10.0	10.7	11.3	12.1	12.5	13.7	14.6
B5. Combination of B1-B4 using one-half standard deviation shocks	9.2	11.8	15.6	16.6	18.0	18.3	18.7	18.6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	9.2	13.7	14.4	14.9	15.7	16.3	18.4	20.3
NPV of debt-to-exports ratio								
Baseline	80.8	79.6	84.3	88.8	98.6	98.0	97.9	107.1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	80.8	80.9	87.0	94.0	107.2	109.2	122.3	159.3
A2. New public sector loans on less favorable terms in 2007-26 2/	80.8	84.8	94.5	103.9	119.4	122.1	135.5	165.3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	80.8	79.6	84.3	88.8	98.6	98.0	97.9	107.1
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	80.8	106.7	142.7	162.4	186.6	181.1	163.0	160.5
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008-09	80.8	79.6	84.3	88.8	98.6	98.0	97.9	107.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	80.8	82.3	88.5	95.1	107.5	106.0	102.7	108.6
B5. Combination of B1-B4 using one-half standard deviation shocks	80.8	110.6	154.9	176.3	203.0	197.5	178.7	175.7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	80.8	79.6	84.3	88.8	98.6	98.0	97.9	107.1
NPV of debt-to-revenue ratio								
Baseline	48.6	52.5	54.3	55.3	57.8	60.0	67.8	75.0
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	48.6	62.5	66.8	71.1	76.6	81.5	103.3	136.0
A2. New public sector loans on less favorable terms in 2007-26 2/	48.6	55.9	60.9	64.7	69.9	74.8	93.8	115.8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	48.6	57.7	65.7	73.7	85.1	97.6	164.7	182.2
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	48.6	60.3	72.6	74.2	77.8	78.8	80.3	79.9
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008-09	48.6	53.3	56.1	58.2	61.8	64.2	72.5	80.2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	48.6	54.2	57.0	59.2	63.0	64.9	71.1	76.1
B5. Combination of B1-B4 using one-half standard deviation shocks	48.6	64.1	82.9	87.0	93.7	95.2	97.5	96.9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	48.6	74.1	76.6	78.1	81.6	84.7	95.7	105.9

Appendix Table 2 (Concluded). Benin: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(Percent)

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
Debt service-to-exports ratio								
Baseline	6.0	5.5	5.3	5.5	6.0	6.4	6.3	6.7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	6.0	5.6	5.4	5.7	6.3	6.9	7.4	9.3
A2. New public sector loans on less favorable terms in 2008-27 2/	6.0	5.5	4.4	4.8	5.5	5.5	7.1	10.3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	6.0	5.5	5.3	5.5	6.0	6.4	6.3	6.7
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	6.0	6.6	7.5	9.2	10.5	11.0	11.0	10.5
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008-09	6.0	5.5	5.3	5.5	6.0	6.4	6.3	6.7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	6.0	5.5	5.4	5.7	6.2	6.7	6.7	6.9
B5. Combination of B1-B4 using one-half standard deviation shocks	6.0	6.6	7.6	9.6	11.0	11.4	11.9	11.4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	6.0	5.5	5.3	5.5	6.0	6.4	6.3	6.7
Debt service-to-revenue ratio								
Baseline	3.6	3.7	3.4	3.5	3.5	3.9	4.3	4.7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	3.6	4.3	4.2	4.3	4.5	5.1	6.2	8.0
A2. New public sector loans on less favorable terms in 2008-27 2/	3.6	3.7	2.8	3.0	3.2	3.4	4.9	7.2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	3.6	4.0	4.1	4.6	5.2	6.4	10.6	11.5
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	3.6	3.7	3.8	4.2	4.4	4.8	5.4	5.2
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2008-09	3.6	3.7	3.5	3.6	3.8	4.2	4.7	5.1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	3.6	3.7	3.5	3.5	3.6	4.1	4.6	4.8
B5. Combination of B1-B4 using one-half standard deviation shocks	3.6	3.8	4.1	4.8	5.1	5.5	6.5	6.3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	3.6	5.2	4.8	4.9	5.0	5.5	6.1	6.7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	29.1	29.1	29.1	29.1	29.1	29.1	29.1	29.1

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt-creating flows.
2/ Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and foreign direct investment.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Appendix Table 3. Benin: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004–27
(Percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt 1/	35.8	38.9	14.7			18.2	19.0	19.9	20.6	21.4	22.1		25.3	29.8	
Of which : Foreign-currency denominated	33.8	36.9	12.7			14.2	14.7	15.3	15.7	16.2	16.6		18.3	19.8	
Change in public sector debt	-2.7	3.1	-24.2			3.5	0.8	0.9	0.7	0.8	0.7		0.6	0.4	
Identified debt-creating flows	-0.6	6.3	-26.5			2.5	1.4	1.1	1.1	1.0	1.0		0.8	0.4	
Primary deficit	3.3	4.3	2.3	2.4	1.3	3.3	2.3	2.2	2.4	2.4	2.5	2.5	2.5	2.5	2.5
Revenue and grants	16.4	16.5	16.8			19.0	19.7	20.1	20.2	20.2	20.2		20.2	20.2	
Of which: grants	0.0	0.0	0.0			0.0	1.2	1.3	1.0	1.0	1.0		1.0	1.0	
Primary (noninterest) expenditure	19.8	20.8	19.1			22.2	22.0	22.3	22.6	22.6	22.6		22.6	22.7	
Automatic debt dynamics	-3.9	2.1	-5.8			-0.8	-0.9	-1.1	-1.3	-1.4	-1.5		-1.7	-2.1	
Contribution from interest rate/growth differential	-0.9	-2.1	-2.2			-0.6	-0.9	-1.1	-1.2	-1.3	-1.4		-1.6	-2.0	
Of which : contribution from average real interest rate	0.2	-1.1	-0.8			0.0	0.1	0.0	-0.1	-0.2	-0.2		-0.2	-0.3	
Of which : contribution from real GDP growth	-1.2	-1.0	-1.4			-0.6	-0.9	-1.0	-1.1	-1.2	-1.2		-1.4	-1.7	
Contribution from real exchange rate depreciation	-3.0	4.2	-3.6			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		...	...	
Other identified debt-creating flows	0.0	0.0	-23.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	-23.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-2.1	-3.2	2.3			1.1	-0.6	-0.2	-0.4	-0.2	-0.2		-0.2	0.0	
NPV of public sector debt	10.7	9.9	15.4			14.5	14.9	16.6	17.6	19.0	20.1		25.2	33.7	
Of which : foreign-currency denominated	8.7	36.9	12.7			9.2	9.7	10.2	10.6	11.1	11.5		13.0	14.4	
Of which : external	8.7	36.9	12.7			9.2	9.7	10.2	10.6	11.1	11.5		13.0	14.4	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	2.6	2.1	2.2			0.0	1.5	1.7	1.3	1.3	1.3		1.3	1.3	
NPV of public sector debt-to-revenue and grants ratio (percent)	64.9	59.7	91.7			76.3	75.6	82.6	87.3	94.1	99.5		125.1	167.4	
NPV of public sector debt-to-revenue ratio (percent)	64.9	59.7	91.7			76.3	80.5	88.5	92.0	99.2	104.8		131.8	176.4	
Of which : external 3/	52.7	223.2	75.6			48.6	52.5	54.3	55.3	58.0	60.2		68.1	75.3	
Debt service-to-revenue and grants ratio (percent) 4/	2.0	1.8	4.1			4.1	4.5	3.8	3.5	3.2	3.4		3.8	4.4	
Debt service-to-revenue ratio (percent) 4/	2.0	1.8	4.1			4.1	4.8	4.0	3.6	3.4	3.5		4.0	4.6	
Primary deficit that stabilizes the debt-to-GDP ratio	6.0	1.2	26.5			-0.3	1.5	1.4	1.7	1.6	1.8		1.9	2.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (percent)	3.1	2.9	3.8	4.4	1.1	4.2	5.4	5.7	5.7	6.0	6.0	5.5	6.0	6.0	6.0
Average nominal interest rate on forex debt (percent)	2.1	2.1	0.9	1.9	0.3	3.0	2.3	2.2	2.1	2.1	2.1	2.3	2.1	1.9	2.1
Average real interest rate on domestic currency debt (percent)	-21.2	-2.2	-2.6	-7.2	10.7	0.4	3.3	1.4	0.1	-0.9	-1.7	0.4	...	...	-2.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.3	13.1	-10.3	0.2	12.4	-1.9	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, percent)	0.3	5.4	3.1	2.9	1.8	3.0	2.8	2.8	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Growth of real primary spending (deflated by GDP deflator, percent)	1.8	8.3	-4.7	-79.3	271.6	21.3	4.3	7.2	6.9	6.1	6.1	8.7	5.9	6.0	6.0
Grant element of new external borrowing (percent)	...	...	...	…	…	29.8	29.7	29.7	29.7	30.1	30.2	29.9	30.7	31.6	...

Sources: Beninese authorities; and Fund staff estimates and projections.
1/ Public and publicly guaranteed debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium- and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Appendix Table 4. Benin: Sensitivity Analysis for Key Indicators of Public Debt, 2007–27

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	14	15	17	18	19	20	25	34
A. Alternative scenarios								
A1. No reform 1/	16	17	20	22	25	27	35	45
A2. Primary balance is unchanged from 2007	17	19	22	24	27	29	40	56
A3. Permanently lower GDP growth 2/	17	17	19	20	22	23	32	47
A4. Increasing domestic and regional borrowing 3/	15	18	19	20	21	22	27	35
B. Bound tests								
B1. Continuous volatility cotton sector 4/	17	18	21	24	27	30	42	60
B2. Primary balance is at hist aver. minus 1 std dev. in 2008–09	17	19	23	24	26	27	33	42
B3. Combination of B1-B2 using one-half standard deviation shocks	17	19	23	24	25	26	31	40
B4. One-time 30 percent real depreciation in 2008	17	21	22	23	24	25	30	40
B5. 10 percent of GDP increase in other debt-creating flows in 2008	17	21	21	21	21	22	22	24
NPV of Debt-to-Revenue Ratio 5/								
Baseline	76	76	83	87	94	99	125	167
A. Alternative scenarios								
A1. No reform 1/	83	89	102	110	121	131	170	218
A2. Primary balance is unchanged from 2007	88	94	107	119	132	144	198	279
A3. Permanently lower GDP growth 2/	88	87	93	99	108	116	157	235
A4. Increasing domestic and regional borrowing 3/	78	93	95	100	105	110	135	176
B. Bound tests								
B1. Continuous volatility cotton sector 4/	88	92	106	119	134	147	209	299
B2. Primary balance is at hist aver. minus 1 std dev. in 2008–09	88	99	116	121	127	133	162	208
B3. Combination of B1-B2 using one-half standard deviation shocks	88	97	113	117	123	129	155	199
B4. One-time 30 percent real depreciation in 2008	88	108	111	113	119	124	149	197
B5. 10 percent of GDP increase in other debt-creating flows in 2008	88	105	104	104	106	107	111	121
Debt Service-to-Revenue Ratio 5/								
Baseline	4	5	4	3	3	3	4	4
A. Alternative scenarios								
A1. No reform 1/	4	5	5	5	5	5	7	10
A2. Primary balance is unchanged from 2007	4	6	6	6	6	7	11	15
A3. Permanently lower GDP growth 2/	4	6	5	5	6	6	9	14
A4. Increasing domestic and regional borrowing 3/	4	6	6	6	6	7	11	16
B. Bound tests								
B1. Continuous volatility cotton sector 4/	4	6	6	6	6	7	11	16
B2. Primary balance is at hist aver. minus 1 std dev. in 2008–09	4	6	6	6	6	7	9	12
B3. Combination of B1-B2 using one- half standard deviation shocks	4	6	6	6	6	7	9	12
B4. One-time 30 percent real depreciation in 2008	4	6	6	6	6	7	10	14
B5. 10 percent of GDP increase in other debt-creating flows in 2008	4	6	8	8	8	9	11	13

Sources: Beninese authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth and primary balance are at historical averages.
2/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
3/ Assumes increaing government T-bills-based-fiscal-deficit financing at regional level leading to an increasing share of domestic debt from 4 percent in 2007 to 16 percent by 2027.
4/ Assumes that real GDP growth at historical average minus one standard deviation (which accounts for instability in both cotton production and cotton farmer pricing).
5/ Revenues are inclusive of grants.